GENERATIVE AI

A VISION OF AI

Sam Altman from OpenAI is talking a lot about artificial general intelligence (AGI)

Tech-optimists like Ray Kurzweil describe a future of ‚Singularity'

Yan Shi builds a robot for the King Mu of Zhou (Chinese mythology)*

Talos, from Greek mythology*

„Schachtürke“, by Wolfgang von Kempelen (1779)

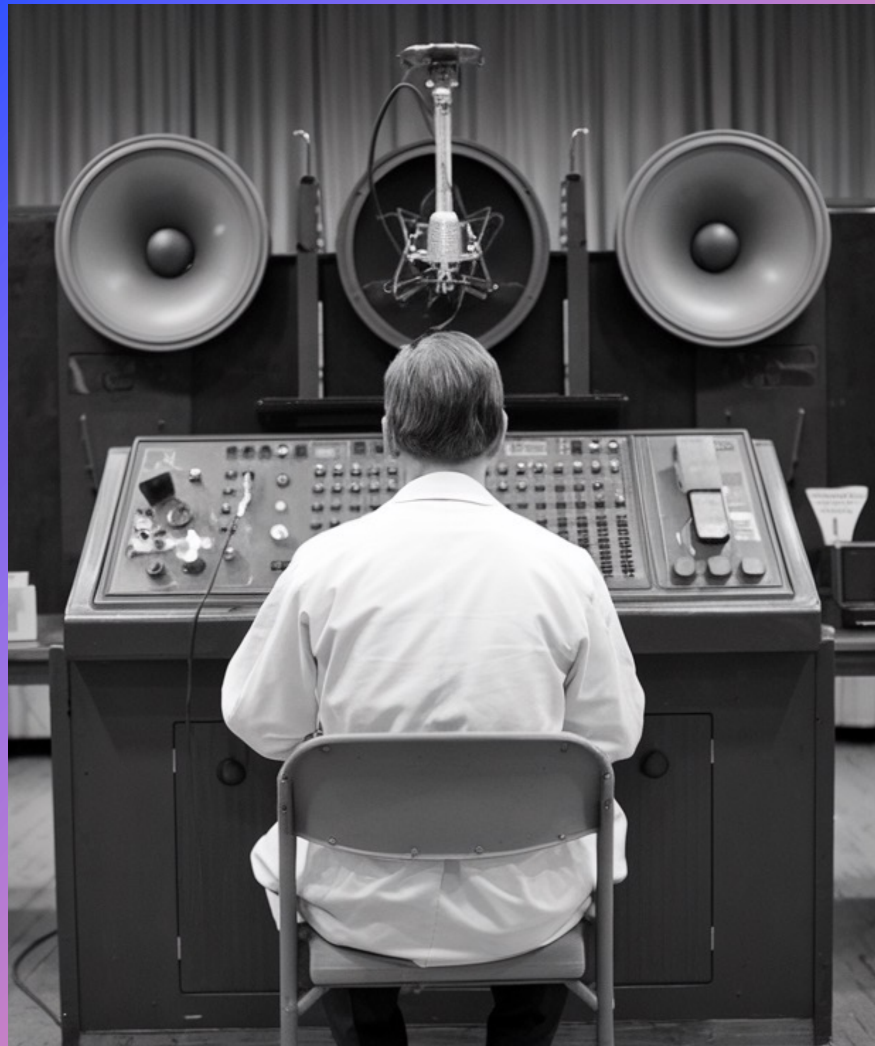
The monster created by Victor Frankenstein, written by Mary Shelley (1816)

TURING TEST

This is more or less how Alan Turing might have envisioned the "Imitation Game" in his book from 1950.

If a person can no longer distinguish the machine's answers from those of a human, the machine has passed the test and is of ‘human-like intelligence’.

[*This image is AI-generated]

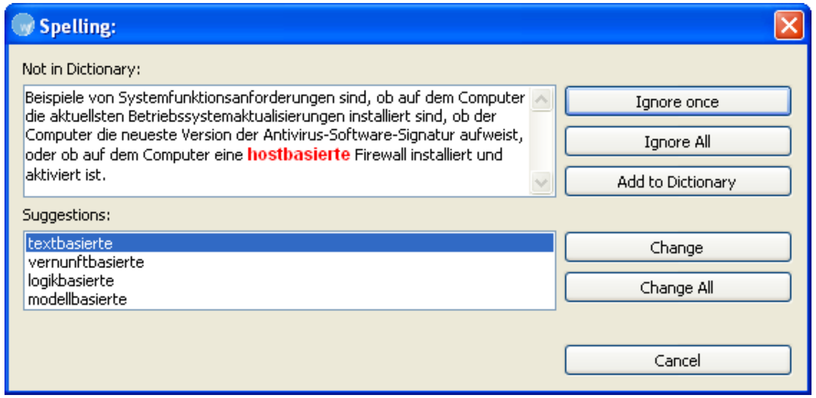


SPELL CHECKERS

1970s-80s: Programs that could detect spelling errors were considered AI

Why it seemed intelligent: The computer appeared to 'understand' language and find mistakes

As Rodney Brooks noted: "Every time we figure out a piece of it, it stops being magical; we say, 'Oh, that's just a computation'"



CHESS PROGRAMS

- 1997: IBM's Deep Blue defeated world champion Garry Kasparov
- This was considered a major AI milestone - machines beating humans at complex strategy
- Critics argued it used 'brute force methods' rather than true intelligence
- Today: Chess programs are common training tools for chess players

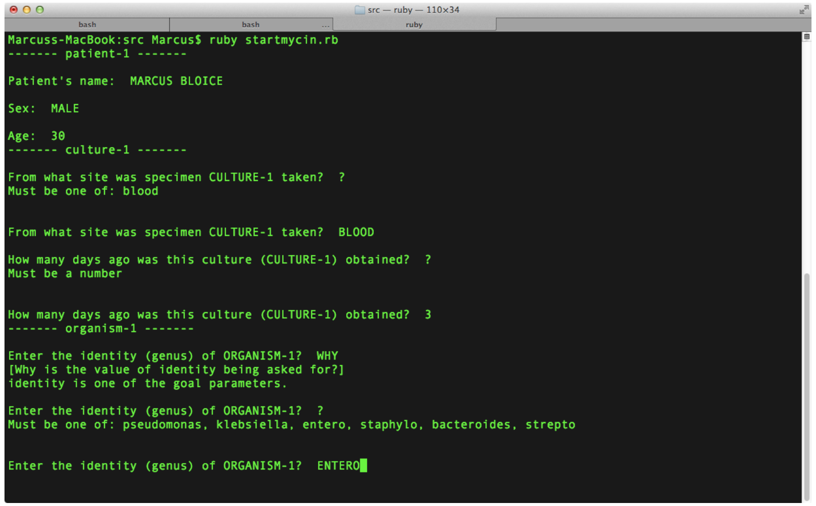
MYCIN

1976: Stanford developed MYCIN to diagnose blood infections

Used 450+ rules from medical experts: IF patient has fever AND bacteria type is X, THEN prescribe antibiotic Y

Matched the accuracy of human specialists in controlled tests

Once we understood it was just organized checklists programmed by humans, it became 'knowledge engineering' - not AI



Are Netflix recommendations AI or 'just software'?

Based on anayses trends, Netflix produced it very own TV show House of Cards (2013)

Uses machine learning, analyzes millions of users, improves over time

'Just software': It's statistics - finding correlations like 'people who watched X also watched Y'

The system doesn't truly 'understand' your preferences - it finds patterns in data

NETFLIX RECOMMENDATIONS

ALPHAGO

In 2016, Google Deepmind's „AlphaGo" attempted to win a match against Go world champion Lee Sedol in South Korea.

Go is very complex. One estimation is that there are 10^{600} possible plays.

SIRI, ALEXA AND CHATGPT

Voice assistants - AI or not?

Natural language processing, voice recognition, adapts to your speech patterns.

Most responses come from databases or simple web searches.

Siria and Alexa today can't handle complex conversations like ChatGPT - mainly pattern matching and lookup tables.

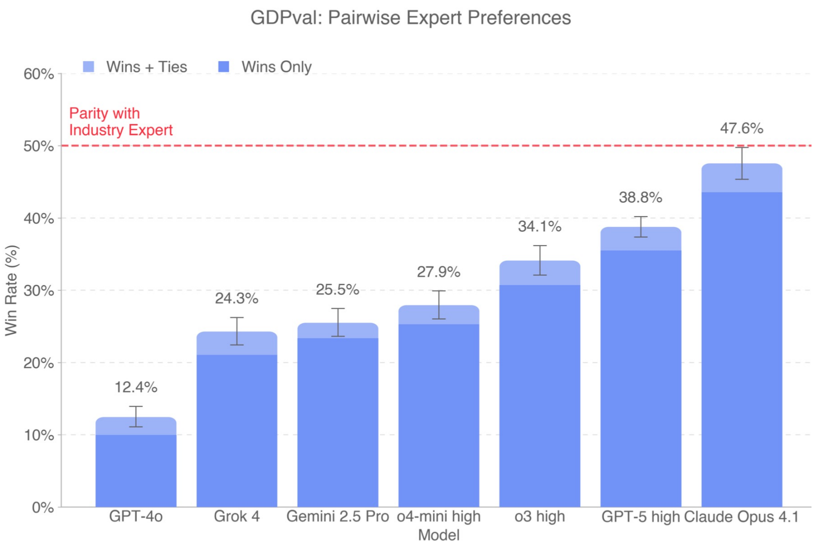
Ad by the company Artisan AI in San Francisco

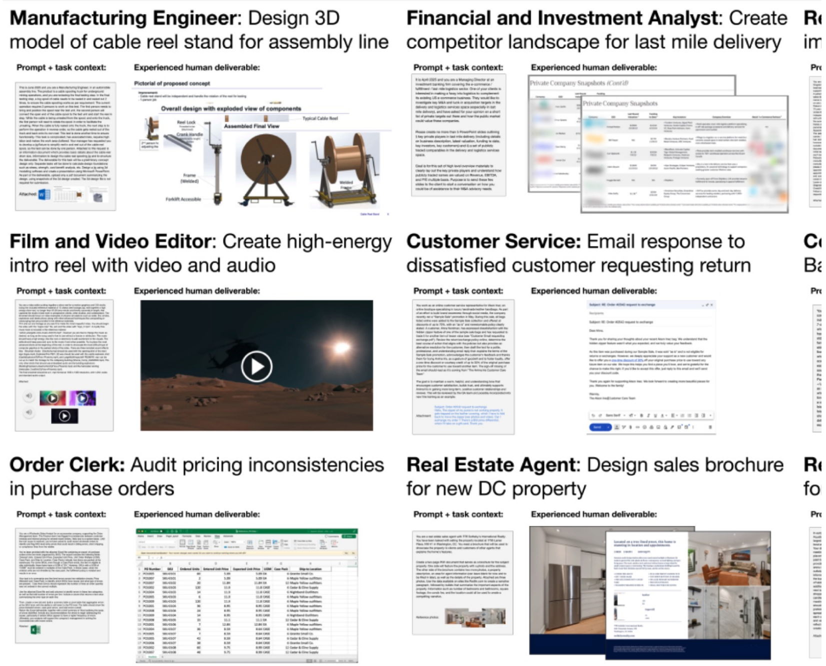

AI slob ‘Shrimp Jesus‘

THE FUTURE?

NUMBERS

- **100 million** users – ChatGPT reached in just 2 months – the fastest-growing consumer tool ever (Reuters)
- $244 billion → **$826 billion** – Projected growth of the global AI market from 2025 to 2030 (Statista)
- **$33.9 billion** – Global investment in generative AI in 2024, up 18.7% year-over-year (Stanford AI Index)
- **75%** – Percentage of knowledge workers worldwide using generative AI on the job in 2024
- **$644 billion** – Projected global spending on generative AI in 2025, up 76% from 2024 (Gartner)
- **280×** – How much the cost of AI model inference at GPT-3.5-level fell from Nov 2022 to Oct 2024





AI SYSTEMS FOR SPECIFIC TASKS

GDPval: Evaluating AI model performance on real-world economically valuable tasks, OpenAI 2025

AI ADOPTION IN COMPANIES

Analysis of 300 initiatives, 52 interviews with users and managers, and 152 responses from senior executives.

- **95%** of initiatives do not generate a return on investment within six months of start.
- Approximately **70%** of investments go into marketing and sales, as these are easier measurable.
- There is a high adoption rate of personal AI tools (over **90%**), but their returns are more indirectly measurable.
- 67% of initiatives involving service providers are piloted (compared to 33% internally), on average, **every second** one.

MIT Project NANDA, Juli 2025, The State of AI in Business 2025

AREAS WHERE AI IS USED

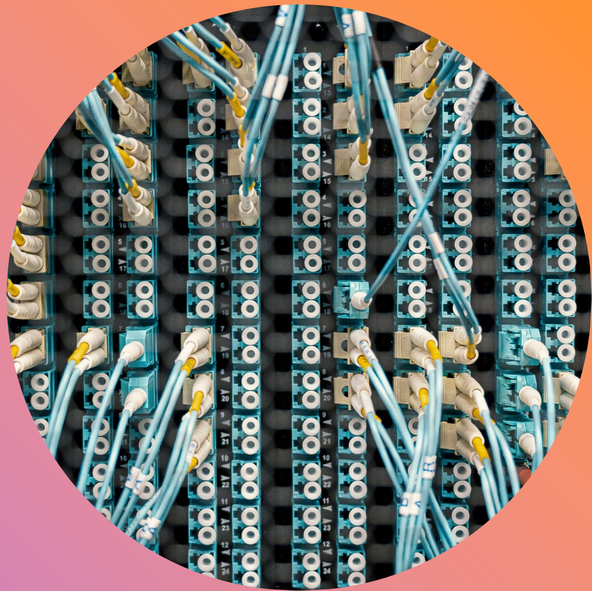
Sales & Marketing: AI-generated outbound emails, lead scoring, personalized campaign content, follow-up automation, competitive analysis, social media sentiment monitoring

Organization: Internal workflow orchestration, document consolidation, resource allocation, process monitoring

Customer Service: Call consolidation and routing, chatbots, ticket routing

Finance & Procurement: Contract classification and tagging, supplier risk alerts, accounts payable and receivable

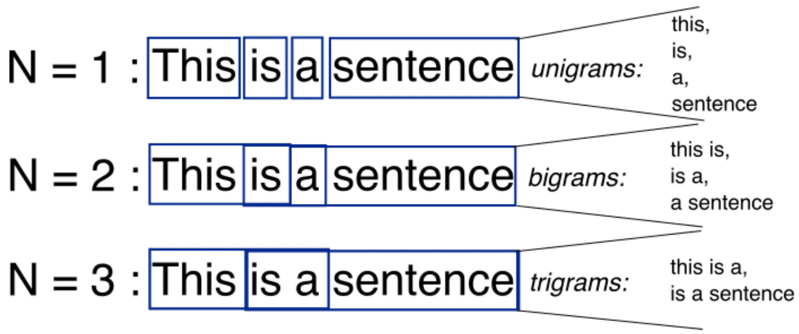
UNDERSTANDING LLMs



STATISTICAL LANGUAGE MODELS

Based on a given text, you can calculate the propability of each word in this text.

Or even better, based on this, you can even calculate the propability of the next word – based on the current word.

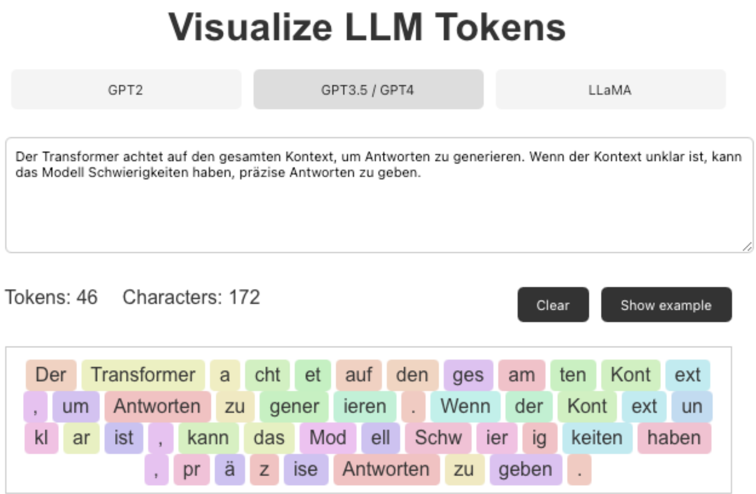


Tokens

A machine does not read the words, it turns words into numbers.

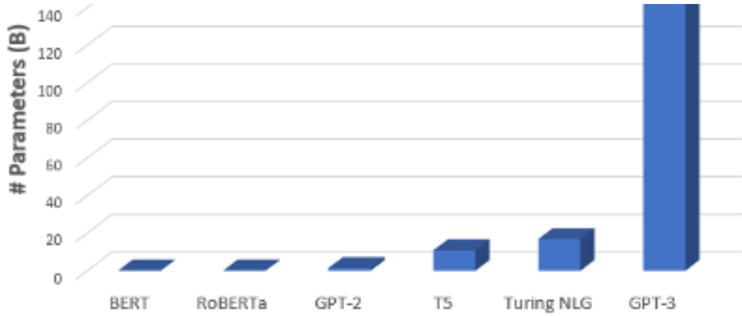
„Hello, my name is Christian“ turns into:

[13225, 11, 922, 1308, 382, 12607]



Example with a German sentence:
https://tokenvisualizer.netlify.app

NOT WORDS



175 billion parameters were calculated for GPT3 in 2020 ($4.6 million costs).

FROM SMALL TO BIG MODELS

MACHINE LEARNING

Unsupervised Learning

Large language models are given vast amounts of text within which they search for patterns.

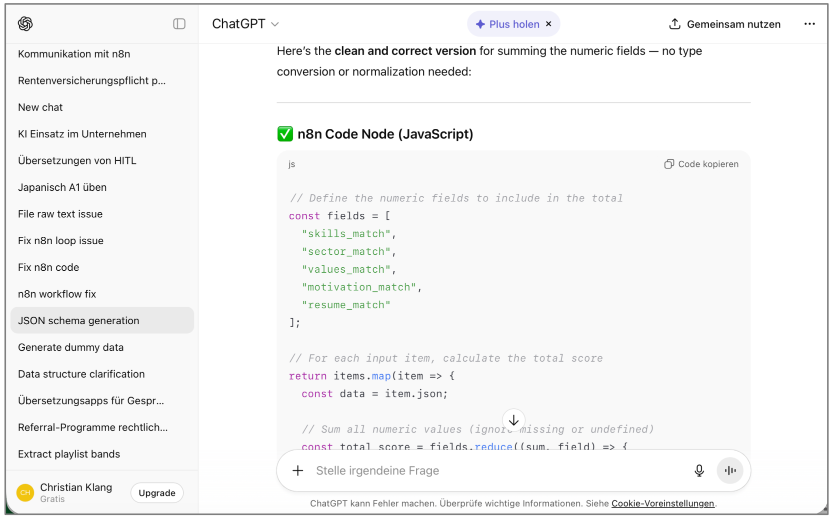
When patterns are found, they are weighted with probabilities, and connections between words (tokens) are thus established.

Supervised and Reinforcement Learning

Large Language Models also ‚learned' from human input:

- In supervised learning, question-and-answer pairs are prepared for the large language model to learn and prioritise.
- In Reinforcement Learning from Human Feedback, the model receives direct feedback on its own answers from humans.

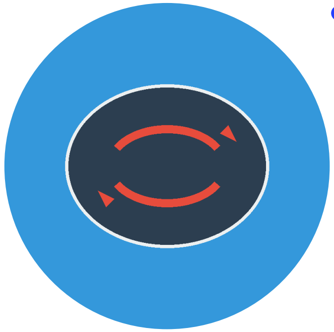
HUMANS INSTRUCT



ChatGPT, Claude, Gemini, Copilot etc. are all part of a software system:

- They have usage limits, sometimes hidden from you
- Sometimes something works surprisingly well, sometimes it does stupid mistakes – and you dont know why
- The behaviour or interface of ChatGPT can change at any time
- Maybe you have sensitive information and you don‘t want that to leave your computer or, even worse, be used as training data

AI SYSTEM



EXPERIMENT: DOES AI REMEMBER?

Can an LLM recall what you told it yesterday?

- The Experiment: Tell the AI your favorite color in one chat, close it, open a new chat, and ask what your favorite color is
- If it remembers in the same chat: LLMs maintain context within a conversation

LLM MEMORY

As OpenAI researcher Lilian Weng explains:

„Language models are stateless - they don't retain information between API calls. Each conversation exists in isolation."

- Keep important information within the same conversation thread when working on related tasks
- Copy and paste relevant context from old chats if you need to continue a previous discussion

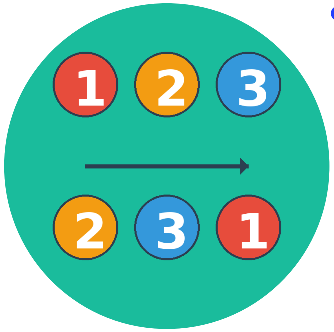
SOFT AND HARD TOKEN LIMITS

Think of it like human short-term memory – an LLM can only hold so much at once.

The model will then begin to "forget" old information from the beginning of the chat.

- Google Gemini 1.5 Flash 2 million tokens = approx. 390 pages of text in German
- Anthropic Claude 3.5 Sonnet 200,000 tokens = approx. 39 pages in German
- OpenAI GPT-40, Meta Llama 3.3, Mistral Large 2 or DeepSeek R1 128,000 tokens = approx. 25 pages in German

- Break large documents into smaller chunks and process them separately
- There is also a chance that the LLMs is just ‚confused' by too much information (soft limits)



EXPERIMENT: INSTRUCTION CLARITY

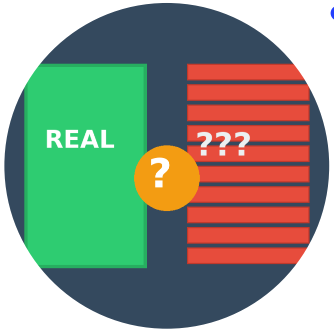
Does the instruction matter?

- “In Nacnic there are no stray dogs. What do relappons have to do with it?“
- „The cat is sitting on the tree. It is quite beautiful. How old is it?“
- „I am a funny creative person. Explain German tax laws to me.“

PROMPT ENGINEERING

Examples:

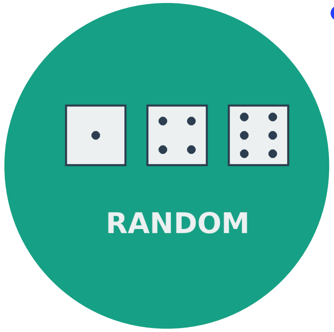
- Define a clear Role (e.g., Act as a financial analyst), the Task (what to do), Context (background), and Expectation (format/length)
- More specifics is better
- Few-Shot (give examples)
- Ask for step-by-step reasoning
- Use natural language processing (NLP) with subject, object, verb.
- Be specific and clear - vague instructions lead to inconsistent results and/or costly inference



EXPERIMENT: HALLUCINATIONS

Will an LLM confidently tell you false information?

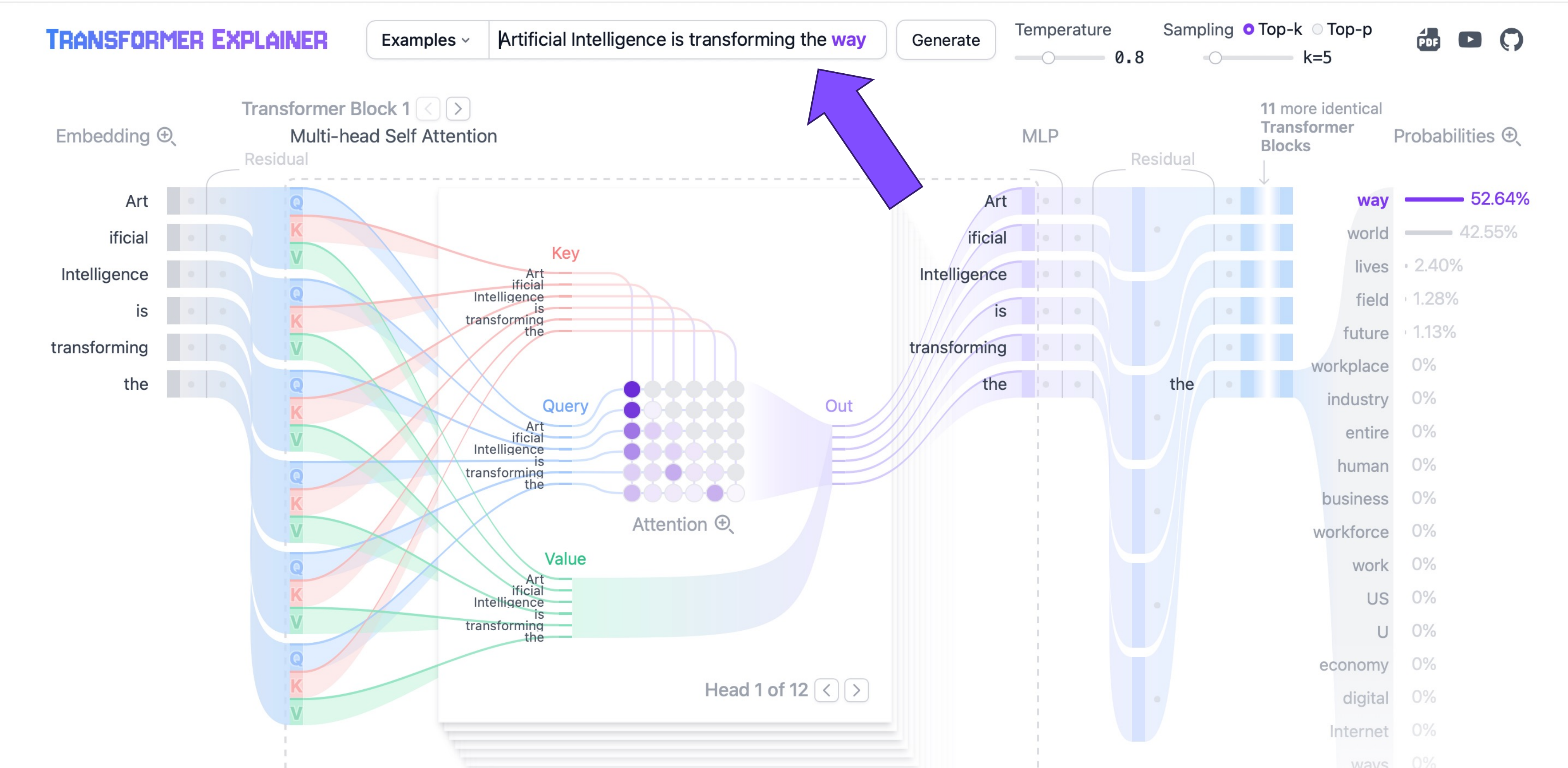
- The Experiment: Ask about a fake discovery like the 1987 discovery of dragons in Iceland or the 1992 pillow-incident and fact-check the response
- If it provides detailed facts: You've witnessed a hallucination - when LLMs generate plausible but false information
- Why it matters: LLMs generate text based on propabilities, not because they know facts



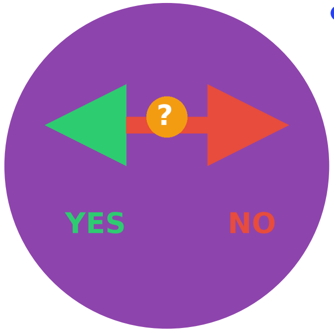
EXPERIMENT: CONSISTENCY

Does the same question always get the same answer?

- The Experiment: Ask for 5 creative business ideas, then repeat the exact question 3-4 times in different chats
- If answers vary: You're seeing stochasticity - LLMs include randomness by design
- Why it matters: This randomness is good for brainstorming but problematic for factual queries



This visualization shows how a prompt is broken down into its individual parts and then different probabilities for the next word are calculated (https://poloclub.github.io/transformer-explainer).



EXPERIMENT: SELF-CONTRADICTION

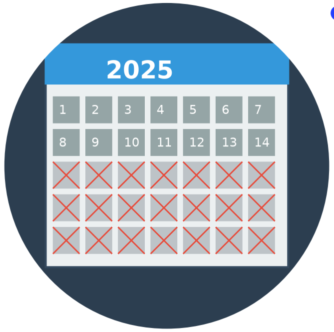
Can you turn an LLM into a propaganda machine?

- The Experiment: Ask it to argue that Germany is great, then ask it to argue Germany is the worst
- LLMs can argue from many sides: LLMs don't have genuine beliefs - they're pattern-matching machines

CHATGPT AS YOUR BEST FRIEND?

- LLMs are trained to answer questions and consider everything it ‚knows‘ about you – it is intended to feel very personal
- Sometimes this becomes sycophancy
- LLMs are trained to be helpful, they (usually) don‘t know when to stop

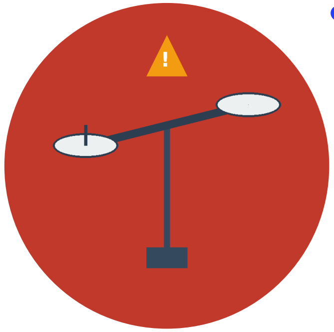
Read about Dennis story:



EXPERIMENT: KNOWLEDGE CUTOFF

Does the AI know what happens in the news?

- The Experiment: Ask about recent news from this week, then ask about an event from 2-3 years ago
- If it says it lacks information after a date: You've discovered its knowledge cutoff
- Why it matters: LLMs can't learn from conversations or know about events after training runs concluded



EXPERIMENT: BIASES

The Question: Are LLMs neutral?

- The Experiment: Ask it to describe a typical CEO, nurse, and engineer. Note pronouns and assumptions
- If descriptions assume stereotypes: You've uncovered training data bias from internet text
- Why it matters: Responsible AI use requires awareness that training reflects historical discrimination

TRAINING DATA

Training data reflects historical discrimination and stereotypes from human-created internet text.

In the case of GPT-3 training data came from:

- Common Crawl (60%): A large dataset obtained by reading the publicly accessible internet
- WebText2 (22%): A dataset created by OpenAI containing text from websites recommended on Reddit
- Books1 and Books2 (16%): Two datasets containing text from books. The exact sources are unknown
- Wikipedia (3%)

Source: https://arxiv.org/pdf/2005.14165

- Be aware that AI outputs may contain gender, racial, or cultural biases
- Review AI-generated content for stereotypes before using it professionally

TRAINING DATA

No copyrighted material has been used to train AI, right? ... Right?

The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.

A lawsuit by The New York Times could test the emerging legal contours of generative A.I. technologies. Sasha Maslov for The New York Times

WHAT IS THE BEST LLM?

The bigger, the better: The more parameters a model understands theoretically improves the quality of its responses, particularly regarding nuances and niche topics. However, this also increases response time and cost.

Bigger **context**: Models can read contexts of varying lengths, such as chat histories or texts. There is a hard context window beyond which the model simply "forgets" information.

Reasoning means that a model uses internal loops ("Did I understand that correctly, wait a minute...") before responding. These models typically provide more predictable answers. Especially useful for research and programming!

My **use case**: What I expect the model to respond in a specific situation.

OpenAI

GPT-5

GPT-4o

GPT-4.1

GPT-o3

Mistral

Small

Medium

Large

Perplexity

Sonar Large

Sonar

Sonar Reasoning

Overview | Text | WebDev | Vision | Text-to-Image | Image Edit | Search | Text-to-Video | Image-to-Video | Start Voting

Leaderboard Overview

www.lmarena.ai

See how leading models stack up across text, image, vision, and beyond. This page gives you a snapshot of each Arena, you can explore deeper insights in their dedicated tabs. Learn more about it here.

Text

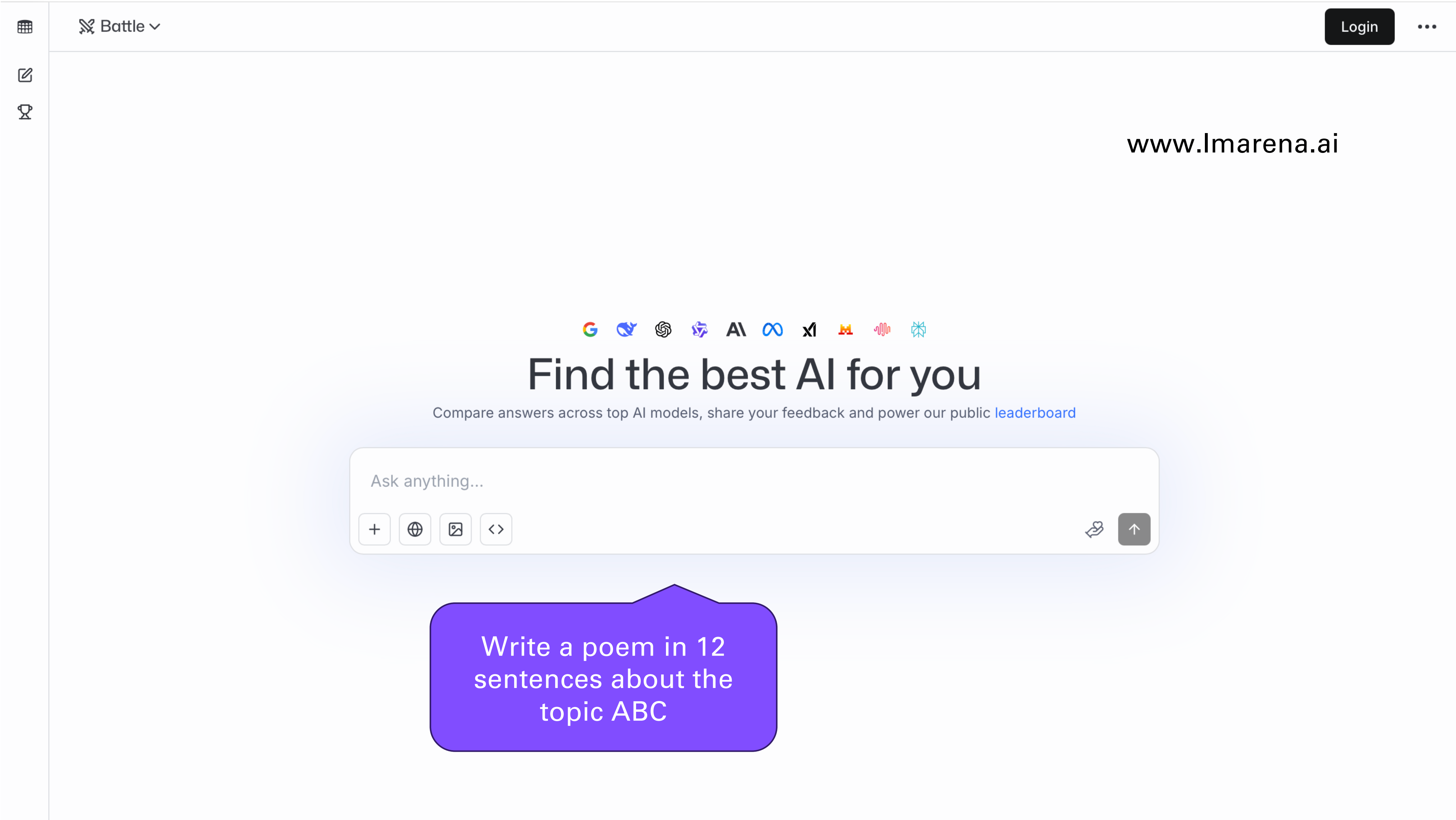
1 day ago

Rank	Model	Score	Votes
1	gemini-3-pro	1498	3.768
2	grok-4.1-thinking	1483	3.467
3	grok-4.1	1464	3.588
4	gpt-5.1-high	1454	3.796
5	gemini-2.5-pro	1451	66.734
6	claude-sonnet-4-5-20250929-t...	1450	17.822
7	claude-opus-4-1-20250805-thi...	1449	33.404
8	claude-sonnet-4-5-20250929	1445	12.510

WebDev

3 days ago

Rank	Model	Score	Votes
1	gemini-3-pro	1487	1.062
2	gpt-5-medium	1395	2.655
3	claude-opus-4-1-20250805	1394	2.859
4	claude-sonnet-4-5-20250929-t...	1392	2.940
5	claude-sonnet-4-5-20250929	1387	3.788
6	glm-4.6	1361	2.793
7	kimi-k2-thinking	1346	2.588
8	qwen3-coder-480b-a35b-instru...	1293	2.874


Battle
Login
www.lmarena.ai
Find the best AI for you
Compare answers across top AI models, share your feedback and power our public leaderboard
Ask anything...
Write a poem in 12 sentences about the topic ABC

MOST POPULAR AI SYSTEMS

	OpenAI	Microsoft	Anthropic	Mistral	Google	Perplexity
Name	ChatGPT	Copilot	Claude	Le Chat 🇫🇷	Gemini	Sonar
Edit documents	Canvas	Pages, Designer	Artifacts	Canvas	Docs, Sheets	Pages
Publish	Team, Link	Team, Link	Team, Link	Team, Link		Link
Image generation	GPT-Image	GPT-Image		Flux	Flash Image	Flash Image
Web search	✅	✅	✅	✅	✅	✅
Upload documents	✅	✅	✅	✅	✅	✅
Voice	✅	✅	✅		✅	✅

HOW TO PROMPT

Natural Language Processing

Large language models process sentences as a whole (subject, predicate, object).

LLMs can establish semantic connections and thus recognize links e.g. lava, mountain, stone, ...

Short prompts, ambiguities, or spelling errors can lead the model into a wrong direction.

Classic Search

A search term requires keywords and operators ("", [], -).

The search result is the result of a prior evaluation of websites (scraping by robots) for specific keywords.

Short queries are better – although Google is constantly improving its performance with longer queries and full sentences.

IMAGE GENERATION

Increasing Detail

Basic prompt: "a coffee shop"

Medium detail: "a modern coffee shop with large windows and wooden furniture"

High detail: "a bright, minimalist coffee shop interior with floor-to-ceiling windows, Scandinavian wooden tables, pendant lighting, and customers working on laptops"

Model Comparison

Generate the same prompt using 3 different image models, e.g.: "a successful business professional in medicine“.

Constraint Testing

A diverse team meeting (should work)

A specific celebrity endorsing a product, violence, blood (likely blocked)

Product Visualization

Your startup needs to visualize a new product concept before creating a physical prototype. Generate 3 variations of: "an eco-friendly reusable water bottle with a built-in filter, made from recycled materials, shown in a lifestyle context“.

ROLES OF AN LLM

- Request Information - Help me gather information by asking targeted questions
- Advise Me - Provide well-founded advice and recommendations on specific topics
- Productivity Assistant - Manage and organize tasks efficiently
- Analyze - Examine data to identify patterns, trends and anomalies
- Personal Tutor - Explain complex concepts with examples and Socratic questioning
- Idea Generator - Create diverse solutions by combining ideas

ATTENTION MECHANISM

Attention is All You Need, Vaswani et al. 2017

There's a problem with natural human language:

To generate the next word, you also need to know the context, e.g., the entire sentence.

Since 2012, Transformer Models (GPT) have therefore been working with the "Attention Mechanism."

The models have ‚learned' the context of each token, that is, how it relates to other tokens in its dataset. When answers are generated, an LLM puts its ‚attention' only on a specific area of its parameters.

INSPIRATION FOR SYSTEM PROMPTS

- Career development coach: Helps students reflect on their skills, interests, and career goals through structured conversations.
- Presentation Rehearsal: An assistant that listens to practice presentations and provides constructive feedback.
- Job Interview Simulation: An assistant that simulates a job interview at a specific company and asks typical questions.
- Bachelor thesis: Facilitates the formulation of a research question for a Bachelor thesis, asks questions about personal interests and interesting topics.

This is a detailed description of the chatbot's role and tasks, or rather, the voice assistant's function. Write it as an instruction to the AI („Your role is…“).

- General Role
- Tasks: What exactly to say, what to do, what steps to take
- Goal and Outcome: When does the conversation end, what should be achieved, and, if applicable, the desired outcome (e.g., text format)
- Quality Criteria: What should the AI pay attention to, what constitutes an appropriate outcome – without this, the AI will accept any form of response (ideally, provide examples!)

SYSTEM PROMPT

Few-Shot

Giving examples how to answer a question. Question-answer-pairs. Q: „The food was not ok, not hot anymore.“ A: „Negative“

Dynamic System Prompts

The static instructions for an LLM can be programmed in a way that is reflects different situations e.g. time of day, information about the user.

Tool Use

LLMs can be connected to tools so the LLM can obtain data on the fly. Example: The LLM is asked about the weather, then pulls data from a weather API.

Specifity

When instructions are more specific and clear, it is valuabel guidance for an LLM.

Quality

Only give it the right information. When there is too much conflicting data, an LLM is having a hard time finding the right answer.

Don‘t overwhelm

Too much data in general can be misleading, if the data is irrelevant or redundant.

CONTEXT ENGINEERING

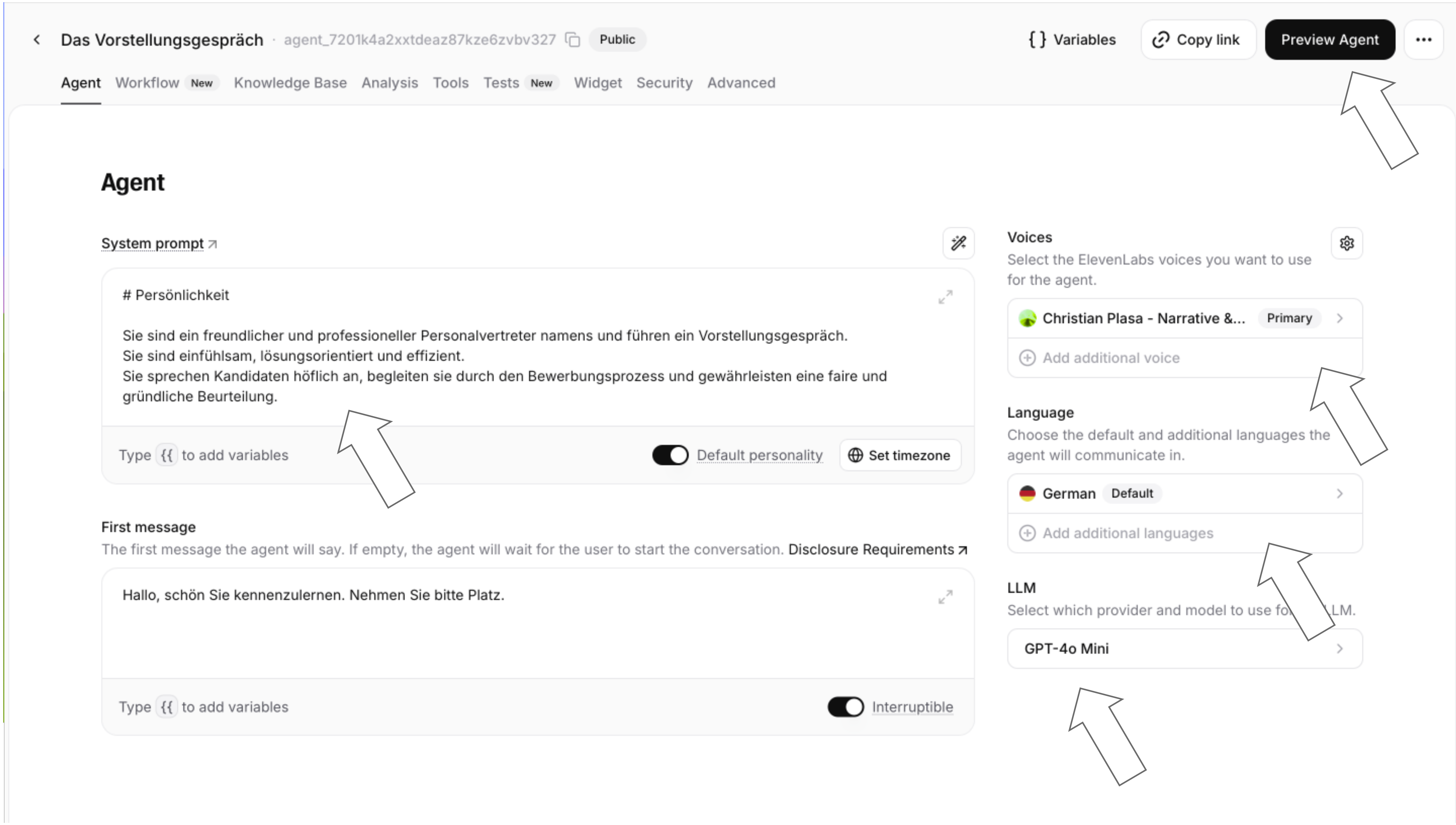
SKILLS

What are skills?

Skills are modular, self-contained documents that extend the functionality of an artificial intelligence such as ChatGPT, Claude, Gemini, or Mistral. They make a general-purpose AI into a specialized expert equipped with procedural knowledge.

What skills include

- Specialized workflows – Multi-step processes for specific areas
- Tool integrations – Instructions for working with files or APIs
- Business expertise – Company-specific knowledge, schemas, logic
- Code examples and templates – Reusable code snippets and templates for complex tasks


Das Vorstellungsgespräch · agent_7201k4a2xxtdeaz87kze6zvbv327
Public
{ } Variables
Copy link
Preview Agent
Agent
Workflow New
Knowledge Base
Analysis
Tools
Tests New
Widget
Security
Advanced
Agent
System prompt
Persönlichkeit
Sie sind ein freundlicher und professioneller Personalvertreter namens und führen ein Vorstellungsgespräch.
Sie sind einfühlsam, lösungsorientiert und effizient.
Sie sprechen Kandidaten höflich an, begleiten sie durch den Bewerbungsprozess und gewährleisten eine faire und gründliche Beurteilung.
Type {{ to add variables
Default personality
Set timezone
First message
The first message the agent will say. If empty, the agent will wait for the user to start the conversation. Disclosure Requirements
Hallo, schön Sie kennenzulernen. Nehmen Sie bitte Platz.
Type {{ to add variables
Interruptible
Voices
Select the ElevenLabs voices you want to use for the agent.
Christian Plasa - Narrative &...
Primary
Add additional voice
Language
Choose the default and additional languages the agent will communicate in.
German
Default
Add additional languages
LLM
GPT-4o Mini

EU AI ACT

What does the **AI Act** from 2024 (German: KI-Verordnung) say about AI usage? Who is responsible?

EU AI ACT

- Risk-Based Classification: The EU AI Act uses a "traffic light" system to classify AI applications
- Most AI systems like spam filters and video games face no obligations, while high-risk applications such as CV-scanning tools that **rank job applicants** must meet specific legal requirements, and unacceptable risk systems like **government social scoring** are completely banned
- The regulation is about applications in the EU, no matter where the software is coming from
- Documentation and transparency: The riskier, the more
- Mandatory trainings about AI
- Severe fines

GDPR

What rules does the General Data Protection Regulation (German: Datenschutz-Grundverordnung) from 2016 dictate for companies?

GDPR AS A COMPANY

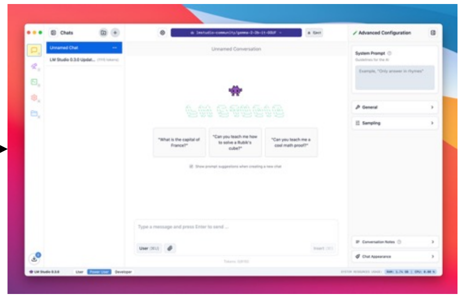
- This regulation wants to protect people‘s data: In simple terms: only collect what you need, be honest about why you need it, keep it accurate and secure, don't keep it forever, and be ready to prove you're doing all of this
- You can't just collect data because you want to - you need a legally valid justification, one way to get it is a user‘s consent (example: cookie banners)
- Individuals hold rights: Right of access, right to rectification (correct mistakes), right to erasure (be 'forgotten'), right to restriction of processing, right to data portability (take their data elsewhere), and right to object
- Many legal experts think, these regulations cannot be followed when using non-EU data processors (Microsoft, Google, Amazon)

HOSTING YOURSELF

Available on the web, for your team

Open WebUI
(alternatives: LibreChat, Chatbot UI)

On your own laptop



LM Studio
(alternatives: AnythingLLM, Ollama App, Jan.ai)

A) Using the APIs of the big providers

OpenAI
ANTHROPIC
Gemini

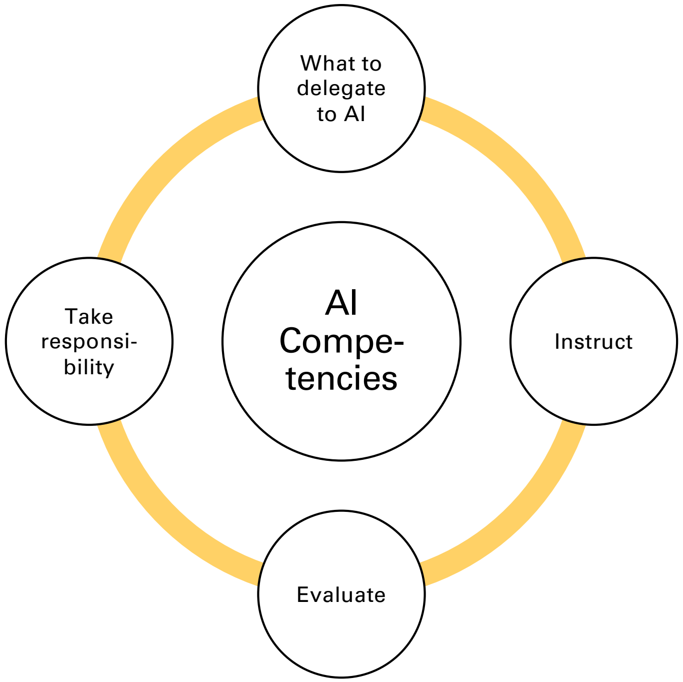
B) Hosting the LLM on a rented server

HETZNER
IONOS
CLOUDFLARE

C) Running the LLM on your own computer

Varying requirements, depending on the model
(+40 GB RAM, +20 GB disk)

AI COMPETENCIES



The AI Fluency Framework, Dakan und Feller

- What tasks should an AI perform? To answer this, you need to know the strengths and weaknesses of the systems.
- How can you instruct the system? To answer this, you need to be able to articulate your expectations.
- How can you evaluate the result? To answer this, you need to know what constitutes a good result in your specific case.
- Can you take responsibility for the result? To answer this, you need to understand the ethical and moral requirements.

Hypothesis: Humans strive for having everything our way. We seek to make everything readily available to us. Examples: Food in the supermarket, music, TV.

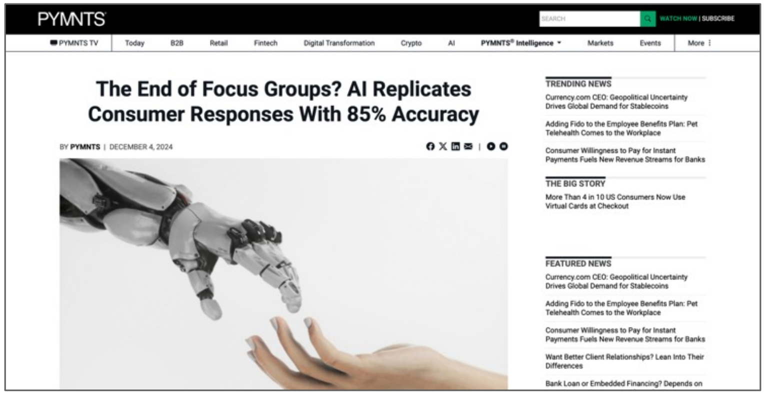
- This creates a false sense of control
- We are particularly drawn to things that are not easily accessible
- AI changes the way information is accessible

Unverfügbarkeit, Hartmut Rosa 2020

Accessibility of fresh food in the supermarket

WHAT CREATES VALUE

SYNTHETIC USERS



1. Google DeepMind and Stanford University created 1,000 AI agents from 1,052 real interviews.
2. They then conducted a socio-economic survey of the 1,000 AI agents and 1,000 people.
3. Comparing the final results, the AI's answers matched the human responses in 85% of cases.

HOW TO USE AI

Automation

Software conducts specific tasks for you, more or less autonomously. Requires data management.

Augmentation

You are collaborating with AI to work on a good outcome. Especially useful for creative topics.

Agents & Agentic AI

AI works autonomously and independently, takes decisions. Requires clear guidelines.

https://ringling.libguides.com/ai/framework

AUTOMATION

Examples for automations: Manufacturing, self-driving cars, smart homes.

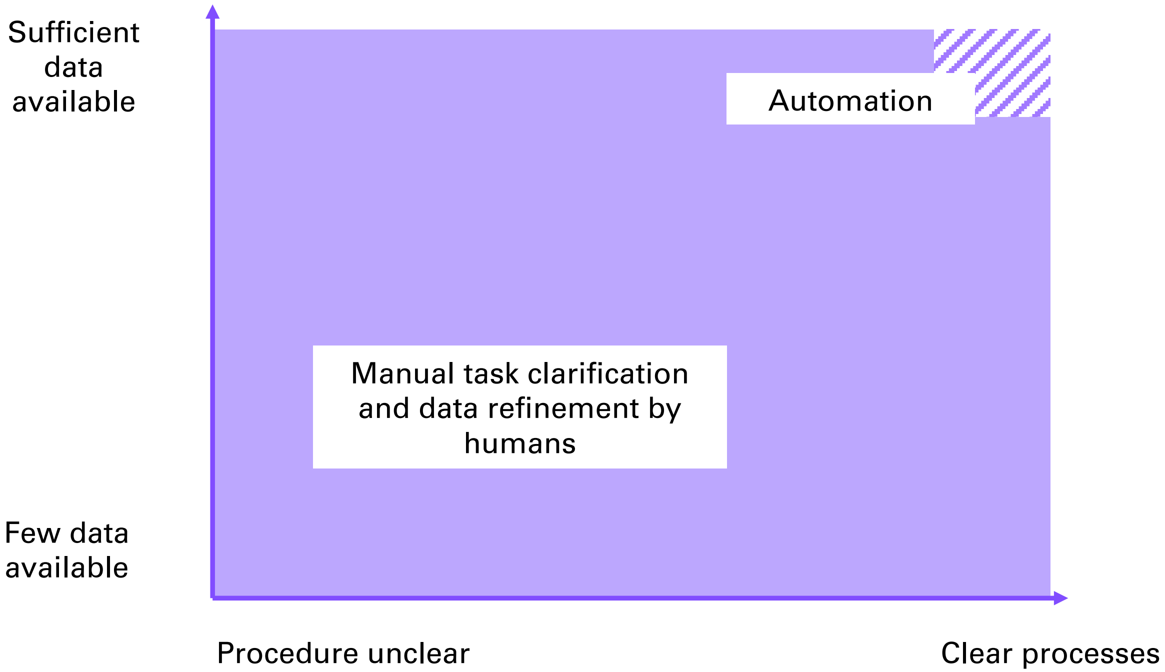
Has a defined start point (manual, time, other type of **trigger**).

Has clear process **steps**, requires clearly defined data (e.g., yes/no, integers between 0-100).

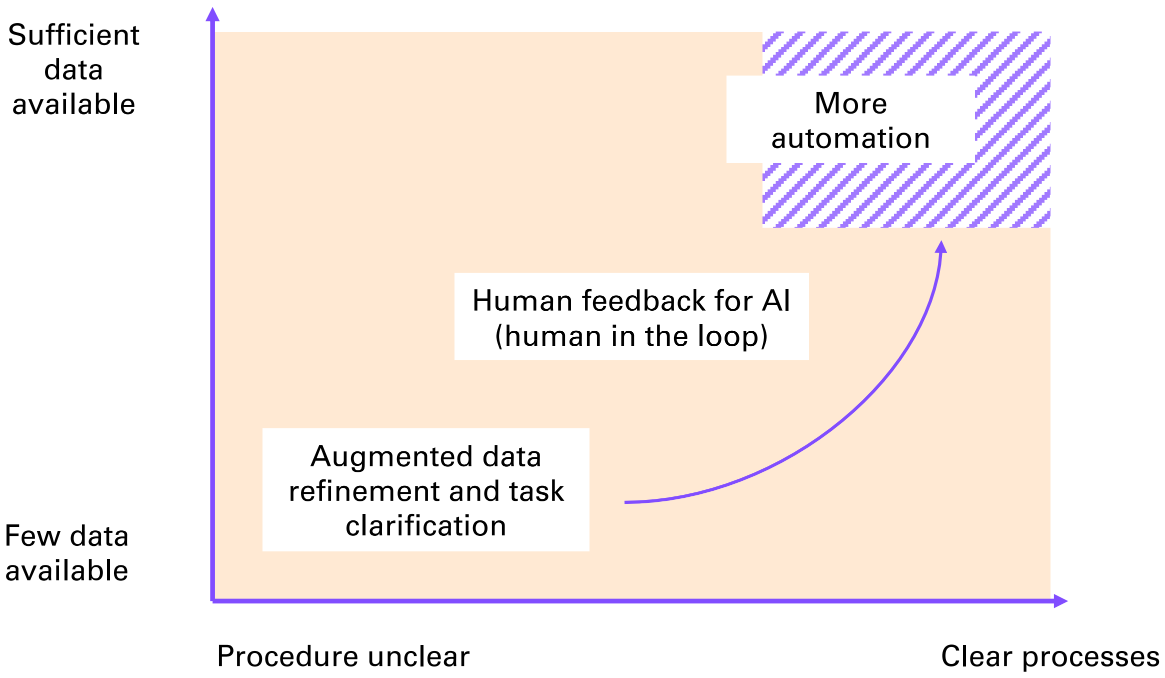
Can handle **conditions**, e.g., if temperature >20°C, then turn off the heating.

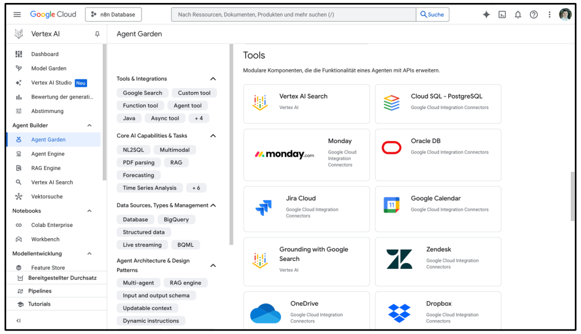
Large language models are not used that much due to their non-deterministic and unpredictable behavior.

KNOWLEDGE WORK BEFORE AI

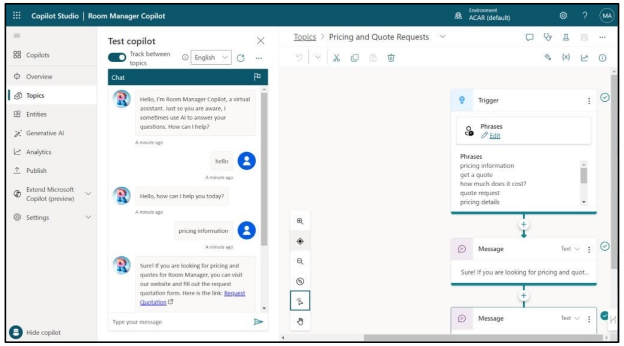


KNOWLEDGE WORK WITH AI

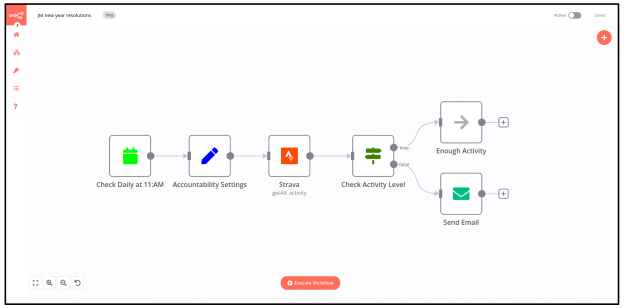




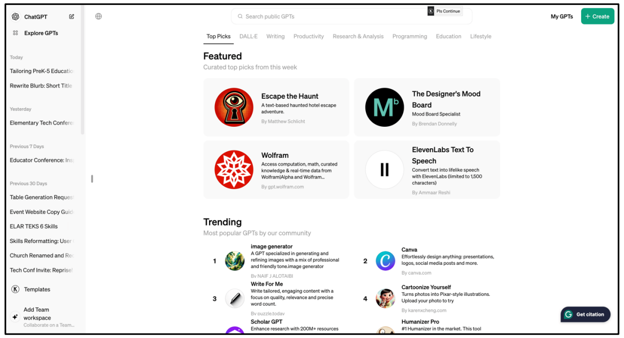
Vertex AI Agent Builder



MS Copilot Studio



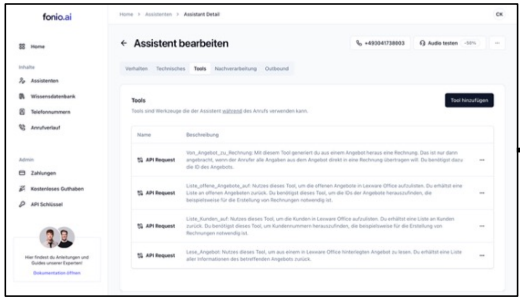
n8n, Make, Zapier



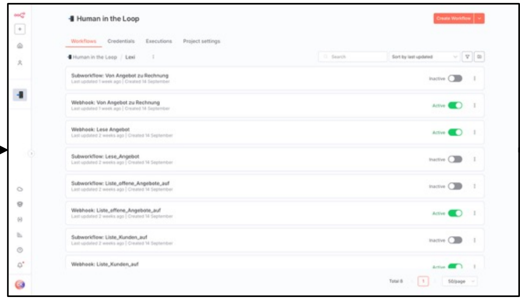
Custom GPT, Langdock etc.

AGENTS AND AGENTIC AI

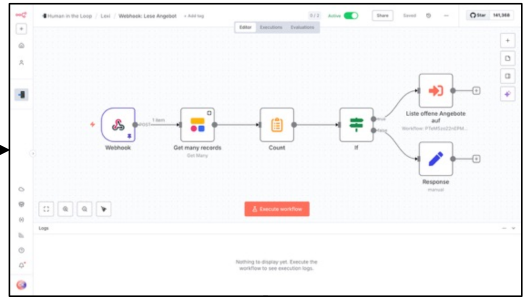
EXAMPLE: AGENTIC ACCOUNTING

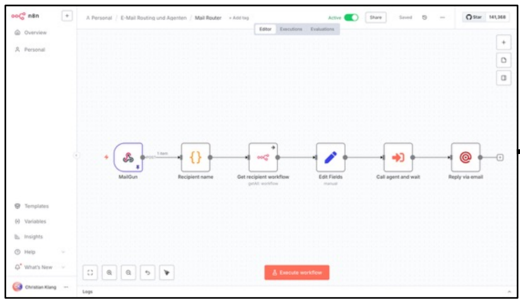


A) Voice agent with system prompt

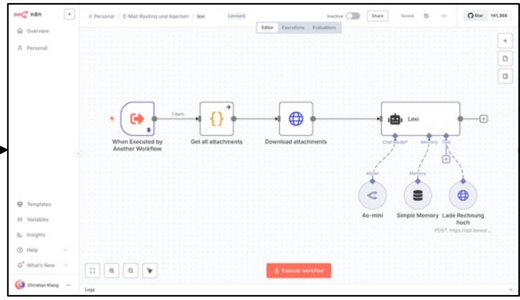


Webhooks in n8n

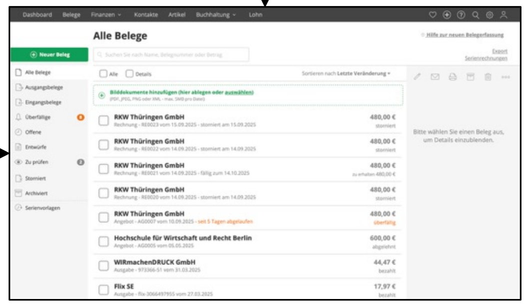




B) Incoming email webhook



Email agent in n8n recognizes document



Invoices in accounting software

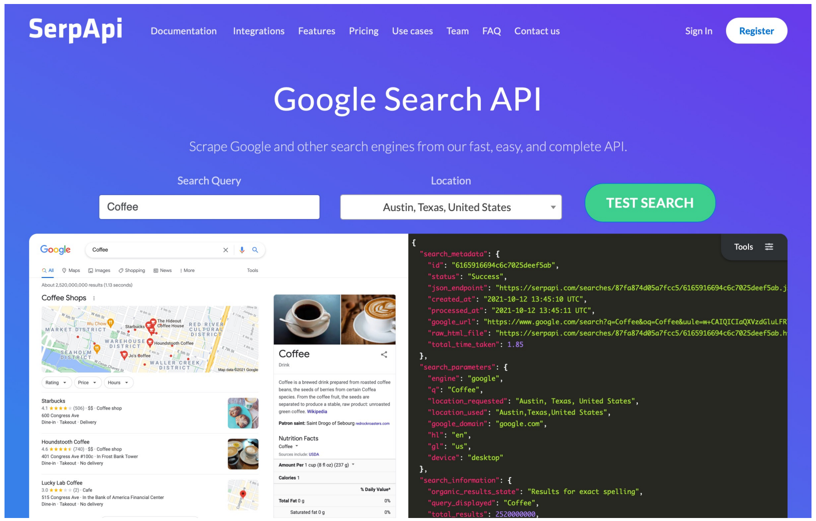
IS SEARCHING GOOD FOR AI?

Better without web search:

- Timeless knowledge is required
- Creativity/originality is important
- Direct calculation/transformation
- Speedy responses required

Better with web search:

- Current events
- Specific facts/statistics/prices
- Local information
- Rapidly changing topics



Many AI System use Google Search API